

Software Test Plan and Test Cases

Online Examination Portal — Phase 1 | IEEE-style Test Plan

Team E6

Document Control

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1. Introduction

1.1 Purpose

This Test Plan defines the Phase-1 validation strategy, test approach, security validation, functional validation, non-functional validation, penetration-testing plan, detailed test cases, and traceability to the SRS.

1.2 Scope

Testing covers authentication, role-based access, question management, examination configuration, question selection, answer handling, submission, evaluation, result release, reporting, security, performance, usability, reliability, compatibility, scalability, data integrity, and auditability.

1.3 Objectives

- Verify coverage of all finalized functional requirements.
- Validate the security objectives and security requirements.
- Validate measurable NFR targets and proposed quality requirements.
- Provide more than the required 7–10 test cases.
- Maintain SRS-to-validation-to-test traceability.

1.4 Test Levels

- Unit testing
- Integration testing
- System testing
- Security testing
- Performance/non-functional testing
- User acceptance validation

2. Test Strategy

- Use positive, negative, boundary, authorization, timing, and data-integrity tests.
- Compare two student attempts to verify the same selected question set and shuffled presentation.
- Validate manual and automatic submission.
- Test result visibility before and after the configured release time.
- Use role-based access tests for Student, Teacher, and Administrator.
- Use performance measurements for measurable response-time targets.
- Penetration testing is planned only; execution is outside the Phase-1 planning deliverable.

3. Test Items and Features

Area	Requirements Covered
Authentication and roles	FR-01, FR-02, FR-03, FR-22, FR-23, FR-24, SEC-01, SEC-02, SEC-03
Question bank	FR-05, FR-06, FR-07
Examination configuration	FR-04, FR-08, FR-21
Question selection/order	FR-09, FR-10, FR-11
Attempt and answers	FR-12, FR-13, FR-14, FR-15, FR-18
Evaluation	FR-16, FR-17
Administrator functions	FR-19, FR-20
Quality attributes	NFR-01 to NFR-12
Security	SEC-01 to SEC-10

4. Test Approach

4.1 General Approach

- Positive tests confirm expected valid workflows.
- Negative tests confirm invalid or unauthorized operations are rejected.
- Boundary tests cover duration, availability, result-release time, and input limits.
- Consistency tests compare behavior across multiple students.
- Security tests verify authentication, authorization, session handling, input validation, and protected data.
- Non-functional tests use measurable acceptance criteria where targets are defined.

4.2 Entry Criteria

- Approved Phase-1 requirements are available.
- Test environment is accessible.
- Required test accounts and examination data are prepared.
- Application build is deployable for the planned validation cycle.

4.3 Exit Criteria

- All planned critical test cases are executed.
- No unresolved critical defect remains for release of the tested Phase-1 scope.
- Requirement-to-test traceability is complete.
- Known limitations and failed tests are documented.

5. Validation Plans

5.1 Security Validation

ID	Area	Validation	Requirement
VAL-S01	Authentication	Verify valid users can authenticate and invalid credentials are rejected.	SEC-01
VAL-S02	Authorization	Verify Student, Teacher, and Administrator permissions are enforced.	SEC-02
VAL-S03	Unauthorized access	Verify a student cannot access another student's attempt/result.	SEC-03
VAL-S04	Input validation	Verify malformed or unexpected input is rejected safely.	SEC-04
VAL-S05	Session security	Verify inactive sessions expire according to policy.	SEC-07
VAL-S06	Question protection	Verify unreleased examination questions cannot be accessed by unauthorized users or exposed before publication.	SEC-09
VAL-S07	Result protection	Verify results are not exposed before release time.	SEC-06
VAL-S08	Privilege escalation	Verify a lower-privileged role cannot invoke administrative operations.	SEC-02
VAL-S09	Sensitive error handling	Verify errors do not expose credentials, stack traces, or sensitive internal details.	SEC-10
VAL-S10	Audit logging	Verify security-relevant administrative actions are recorded.	SEC-08
VAL-S11	Attempt integrity	Verify submitted answers/attempt records cannot be modified by unauthorized users.	SEC-05

5.2 Functional Validation

ID	Area	Validation	Requirement
VAL-F01	Authentication and role access	Valid Student/Teacher login works; invalid credentials are rejected.	FR-01, FR-02, FR-03
VAL-F02	Question upload	Teacher can add an objective question to the question bank.	FR-05
VAL-F03	Difficulty	Teacher can assign a difficulty level.	FR-06
VAL-F04	Exam creation/configuration	Teacher can create an exam and configure duration/availability.	FR-04, FR-08
VAL-F05	Random selection and same set	Two students receive the same selected set for the same exam.	FR-09, FR-10
VAL-F06	Shuffled order/display	Selected questions are displayed in shuffled order and linked to the active attempt.	FR-11, FR-12
VAL-F07	Submission timing	Manual submission works and timeout triggers automatic submission.	FR-14, FR-15

VAL-F08	Evaluation	Objective answers are evaluated and score is calculated correctly.	FR-16, FR-17
VAL-F09	Result release	Result is unavailable before release time and available after release time.	FR-19
VAL-F10	Result timing boundary	Result-release behavior is correct at the configured release boundary.	FR-19
VAL-F11	Available examinations	Student sees only examinations within the configured availability period.	FR-21
VAL-F12	Answer save/change	Student can save, revisit, and change answers before submission.	FR-13
VAL-F13	Pre-publication question management	Teacher can edit/remove questions before publication and is blocked afterward.	FR-07
VAL-F14	Teacher/administrator reports	Authorized teachers and administrators can view examination reports.	FR-20
VAL-F15	User administration	Administrator can manage users.	FR-03
VAL-F16	Role administration	Administrator can manage roles according to authorization.	FR-03
VAL-F17	Examination lifecycle	Exam moves through configured states without invalid transitions.	FR-04, FR-08, FR-14, FR-19
VAL-F18	Administrative access	Unauthorized users cannot access administrative operations.	FR-03
VAL-F19	Attempt persistence	Attempt and answer data persist correctly through submission.	FR-18
VAL-F20	Administrator Login	Valid administrator credentials allow login; invalid credentials are rejected and protected functions remain inaccessible until authentication.	FR-22 / UC-A06
VAL-F21	System settings	Authorized administrator can update approved system-level settings; unauthorized roles are rejected.	FR-23 / UC-A03
VAL-F22	Access management	Authorized administrator can manage defined access permissions; unauthorized roles are rejected.	FR-24 / UC-A05

5.3 Non-Functional Validation

ID	Area	Validation	Requirement
VAL-N01	Security	Measure protected-function access and authorization behavior against NFR-01.	NFR-01
VAL-N02	Reliability	Repeat answer save/submission workflows and verify no loss/corruption of attempt data.	NFR-02
VAL-N03	Performance	Measure common user actions; target is ≤ 3 seconds under normal load.	NFR-03

VAL-N04	Availability	Verify authorized access throughout a configured examination availability period, excluding planned maintenance.	NFR-04
VAL-N05	Usability	Evaluate navigation, instructions, feedback, and task completion.	NFR-05
VAL-N06	Learnability	Observe a first-time student completing start-to-submit within 5 minutes without assistance.	NFR-06
VAL-N07	Maintainability	Review modularity, interface documentation, and change isolation.	NFR-07
VAL-N08	Compatibility	Run core workflows on supported current major desktop browsers.	NFR-08
VAL-N09	Testability	Check that each requirement has an observable validation method and acceptance condition.	NFR-09
VAL-N10	Scalability	Run concurrent-user tests and observe examination workflow stability.	NFR-10
VAL-N11	Data integrity	Compare stored questions, answers, attempts, scores, and results before/after critical operations.	NFR-11
VAL-N12	Auditability	Verify important administrative/examination actions create usable audit records.	NFR-12

6. Test Environment

- Web browser environment for Student, Teacher, and Administrator roles.
- Application server and persistent database environment.
- Test accounts representing each role.
- Seeded objective-question bank and examination configurations.
- Controlled test data for duration, availability, and result-release boundaries.
- Separate security-testing environment for planned penetration testing.

7. Detailed Test Cases

ID	Test Case	Precondition	Steps	Expected Result	Traceability	Type
TC-01	Valid student login	Student account exists	Enter valid credentials	Student dashboard is displayed	FR-01 / VAL-F01	Functional
TC-02	Invalid login	Student account exists	Enter invalid password	Access is rejected with safe feedback	FR-01 / SEC-01	Security
TC-03	Teacher creates exam	Teacher is authenticated	Create exam and configure duration/availability	Exam is created with configured settings	FR-04, FR-08 / VAL-F04	Functional
TC-04	Question upload and difficulty	Teacher is authenticated	Upload objective question and assign difficulty	Question is stored with difficulty	FR-05, FR-06 / VAL-F02,	Functional

					VAL-F03	
TC-05	Same selected question set	Exam is initialized	Start attempts for two students	Both receive the same selected question set	FR-09, FR-10 / VAL-F05	Functional
TC-06	Shuffled order	Exam has selected questions	Start attempts and compare question order	Question order is shuffled as specified	FR-11 / VAL-F06	Functional
TC-07	Save/change answer	Student has active attempt	Save answer, revisit, change answer	Latest answer is retained before submission	FR-13 / VAL-F12	Functional
TC-08	Automatic submission	Exam duration is configured	Allow timer to expire	Attempt is automatically submitted	FR-14 / VAL-F07	Functional
TC-09	Manual submission	Student has active attempt	Submit before timeout	Attempt is submitted successfully	FR-15 / VAL-F07	Functional
TC-10	Automatic evaluation	Submitted objective answers exist	Complete submission	Correct answers are evaluated and score is calculated	FR-16, FR-17 / VAL-F08	Functional
TC-11	Result release timing	Result release time is configured	Check result before and after release	Result is hidden before and visible after release	FR-19 / VAL-F09, VAL-F10	Functional
TC-12	Teacher/administrator or report access	Teacher or administrator has authorized examination/report access	Open authorized examination report	Report is displayed	FR-20 / VAL-F14	Functional
TC-13	Question lock after publication	Exam is published	Attempt to edit/remove associated question	Operation is rejected	FR-07 / VAL-F13	Security
TC-14	Role restriction	Student account is authenticated	Attempt teacher/admin endpoint	Access is denied	FR-03 / VAL-F18, VAL-S02	Security
TC-15	Unauthorized result access	Student A and B exist	Student A requests Student B's result	Request is rejected	SEC-03 / VAL-S03	Security
TC-16	Session expiry	Authenticated session is idle	Wait for configured session timeout and access protected page	Session is expired and re-authentication is required	SEC-07 / VAL-S05	Security
TC-17	Attempt integrity	Submitted attempt exists	Attempt unauthorized modification	Modification is rejected and stored data remains unchanged	FR-18 / VAL-S11	Security
TC-18	Response time	Normal-load test environment	Perform common user actions	Response time is ≤ 3 seconds for the defined actions	NFR-03 / VAL-N03	Non-functional
TC-19	First-time usability	New student account	Start and submit an exam without assistance	Task completes within 5 minutes	NFR-06 / VAL-N06	Non-functional
TC-20	Browser compatibility	Supported browsers available	Run login, exam, submit, and result workflows	Core workflow behaves consistently	NFR-08 / VAL-N08	Non-functional
TC-21	Data integrity	Known exam/attempt data exists	Compare data across save, submit, evaluate, release	No unintended data loss or	NFR-11 / VAL-N11	Non-functional

				inconsistency occurs		
TC-22	Concurrent users	Load environment configured	Run multiple concurrent examination sessions	Core workflow remains stable under target load	NFR-10 / VAL-N10	Non-functional
TC-23	Audit logging	Audit feature enabled	Perform administrative/security-relevant actions	Required audit records are created	NFR-12 / VAL-N12	Non-functional
TC-24	Error information safety	Invalid request available	Trigger malformed/unauthorized request	Safe error response is returned without sensitive details	SEC-04, SEC-10 / VAL-S04, VAL-S09	Security
TC-25	Administrator login	Administrator account exists	Enter valid and invalid administrator credentials	Valid credentials open the administrator dashboard; invalid credentials are rejected	FR-22 / UC-A06 / VAL-F20	Functional/Security
TC-26	Manage system settings	Administrator is authenticated	Open settings and change an approved setting	Setting is updated; unauthorized roles are rejected	FR-23 / UC-A03 / VAL-F21	Functional/Security
TC-27	Manage access permissions	Administrator is authenticated	Update an allowed role permission and attempt unauthorized access	Permission follows defined role rules; unauthorized operation is rejected	FR-24 / UC-A05 / VAL-F22	Security/Functional
TC-28	Teacher login	Teacher account exists	Enter valid teacher credentials	Teacher dashboard is displayed	FR-02 / VAL-F01	Functional
TC-29	Available examinations	Student is authenticated; examinations exist with different availability periods	Open the available-examinations view	Only examinations currently within their configured availability period are displayed	FR-21 / VAL-F11	Functional
TC-30	Examination availability	Authorized user and configured examination availability period exist	Access the portal during the configured availability period and verify service availability	Authorized users can access the system throughout the configured examination availability period, excluding planned maintenance	NFR-04 / VAL-N04	Non-functional
TC-31	User administration	Administrator is authenticated	Create, update, or deactivate an authorized user account	User account change is applied according to authorization rules	FR-03 / VAL-F15	Functional/Security
TC-32	Role administration	Administrator is authenticated	Assign or manage a defined system role	Role change is applied	FR-03 / VAL-F16	Functional/Security

			according to authorization	according to authorization rules		
TC-33	Question confidentiality before publication	Examination contains unreleased questions; unauthorized user is authenticated or has an available endpoint	Attempt to access question content before the examination is published	Question content is not exposed to the unauthorized user	SEC-09 / VAL-S06	Security
TC-34	Usability evaluation	Student account and test examination are available	Navigate through login, available-exam, attempt, answer, submission, and result workflows	Navigation, instructions, feedback, and task completion are clear and understandable	NFR-05 / VAL-N05	Non-functional

8. Traceability to SRS

SRS Requirement	Test Case	Validation
FR-01	TC-01	VAL-F01
FR-02	TC-28	VAL-F01
FR-03	TC-14, TC-31, TC-32	VAL-F01, VAL-F15, VAL-F16, VAL-F18
FR-04	TC-03	VAL-F04
FR-05	TC-04	VAL-F02
FR-06	TC-04	VAL-F03
FR-07	TC-13	VAL-F13
FR-08	TC-03	VAL-F04
FR-09	TC-05	VAL-F05
FR-10	TC-05	VAL-F05
FR-11	TC-06	VAL-F06
FR-12	TC-06	VAL-F06
FR-13	TC-07	VAL-F12
FR-14	TC-08	VAL-F07
FR-15	TC-09	VAL-F07
FR-16	TC-10	VAL-F08
FR-17	TC-10	VAL-F08
FR-18	TC-17	VAL-S11
FR-19	TC-11	VAL-F09, VAL-F10
FR-20	TC-12	VAL-F14
FR-21	TC-29	VAL-F11
NFR-01	TC-14	VAL-N01
NFR-02	TC-21	VAL-N02
NFR-03	TC-18	VAL-N03
NFR-04	TC-30	VAL-N04
NFR-05	TC-34	VAL-N05
NFR-06	TC-19	VAL-N06
NFR-07	Design/code review	VAL-N07
NFR-08	TC-20	VAL-N08
NFR-09	Traceability audit	VAL-N09
NFR-10	TC-22	VAL-N10
NFR-11	TC-21	VAL-N11
NFR-12	TC-23	VAL-N12
FR-22	TC-25	VAL-F20
FR-23	TC-26	VAL-F21
FR-24	TC-27	VAL-F22
SEC-09	TC-33	VAL-S06
SEC-10	TC-24	VAL-S09

SEC-04	TC-24	VAL-S04
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9. Penetration Testing Plan

Penetration testing is planning-only for Phase 1. Execution should be performed in an authorized test environment.

Plan ID	Area	Planned Activity	Requirements
VAL-P01	Authentication testing	Check credential handling, invalid login, session controls.	SEC-01, SEC-07
VAL-P02	Authorization testing	Attempt cross-role and privilege-escalation access.	SEC-02, SEC-03
VAL-P03	Input validation testing	Test malformed and unexpected input at exposed operations.	SEC-04
VAL-P04	Session testing	Check session expiry, reuse, and protected-page access after logout/expiry.	SEC-07
VAL-P05	Data-access testing	Attempt unauthorized access to attempts, answers, questions, and results.	SEC-03, SEC-05, SEC-09
VAL-P06	Result-release testing	Attempt to retrieve results before configured release time.	SEC-06
VAL-P07	Audit and error testing	Verify security events are logged and errors do not disclose sensitive details.	SEC-08, SEC-10

10. Reporting and Completion

- Record tester, date, environment, preconditions, steps, expected result, actual result, pass/fail status, and defect ID for executed cases.
- Classify defects by severity and link them to the affected requirement/test case.
- Re-test corrected defects and update traceability evidence.
- Maintain the final test summary with executed cases, passed/failed cases, open defects, and known limitations.